# Order Book Microstructure & Realized Volatility Prediction

*A Mini Project in High-Frequency Signal Engineering*

**Author:** Anuj Panwar, FRM — **Domain:** Quantitative Finance / HFT Research — **Date:** March 2026

**Abstract.** This mini project constructs a supervised learning pipeline for predicting short-horizon *realized volatility* from Level-1 limit order book (LOB) data. Raw tick data is transformed into three microstructure signals—Weighted Average Price (WAP), bid-ask spread, and order-flow imbalance—which serve as features in a regression framework. The methodology bridges FRM market-risk concepts (realized variance, log-returns, VaR), stochastic calculus (quadratic variation), and practical HFT signal engineering. The project replicates the core modelling task set by the Optiver Realized Volatility Prediction competition on Kaggle.

## 1. Data Source

The dataset originates from the **Optiver Realized Volatility Prediction** competition hosted on **Kaggle** (2021). It provides anonymized, high-frequency Level-1 order book snapshots for a universe of 112 NASDAQ-listed stocks sampled at irregular intraday intervals across multiple trading days.

Each observation in the `book_train.parquet` file corresponds to a single LOB snapshot and contains the following fields:

| Field | Type | Description |
|---|---|---|
| `stock_id` | int | Anonymised stock identifier |
| `time_id` | int | Auction window identifier (10-min bucket) |
| `seconds_in_bucket` | int | Seconds elapsed within the window |
| `bid_price1 / ask_price1` | float | Best bid/ask prices |
| `bid_size1 / ask_size1` | int | Quantities at the best bid/ask |

The target variable—**realized volatility** per 10-minute `time_id` window—is provided in a separate `train.csv` file. Importantly, the data is *publicly available* and can be reproduced via the Kaggle API or downloaded directly from the competition page.

## 2. Pipeline Steps

The pipeline follows five sequential stages:

**Step 1 — Data Ingestion.** Raw Parquet files are loaded using `pandas.read_parquet()`, a columnar format chosen for efficiency with high-frequency tick data (`pyarrow` backend).

**Step 2 — Microstructure Feature Construction.** Three signals are engineered per tick observation:

- **Spread:** $S_t = P_t^{ask} - P_t^{bid}$, the cost of crossing the book.
- **Weighted Average Price (WAP):**

$$\mathrm{WAP}_t = \frac{P_t^{bid} \cdot Q_t^{ask} + P_t^{ask} \cdot Q_t^{bid}}{Q_t^{bid} + Q_t^{ask}}$$

  A size-weighted fair-value estimate that corrects for quote stuffing.

- **Order-Flow Imbalance (OFI):**

$$\mathrm{OFI}_t = \frac{Q_t^{bid} - Q_t^{ask}}{Q_t^{bid} + Q_t^{ask}} \in [-1, 1]$$

A signed measure of buying versus selling pressure.

**Step 3 — Log-Return Computation.** Within each `time_id`, WAP log-returns are computed:

$$r_t = \log\left(\frac{\mathrm{WAP}_t}{\mathrm{WAP}_{t-1}}\right)$$

using `groupby + transform(lambda x: np.log(x).diff())` to preserve group boundaries and avoid look-ahead contamination.

**Step 4 — Realized Volatility Aggregation.** Per window, realized volatility is computed as the square root of the sum of squared log-returns (see Section 3).

**Step 5 — Feature Aggregation & Dataset Construction.** Tick-level features are collapsed to window-level means. The merged dataframe forms the supervised learning dataset: $\mathbf{X} = [\bar{S}, \overline{\mathrm{WAP}}, \overline{\mathrm{OFI}}, \bar{Q}^{bid}, \bar{Q}^{ask}]$ and target $y = \sigma^{RV}$.

## 3. Methodology

### 3.1 Realized Variance and Quadratic Variation

The realized volatility estimator used is the **Andersen–Bollerslev–Barndorff-Nielsen** (ABBN) discrete approximation of quadratic variation:

$$\boxed{\sigma^{RV} = \sqrt{\sum_{i=1}^{N} r_i^2}}$$

where $r_i$ are the intra-window WAP log-returns and $N$ is the number of ticks in the 10-minute window. This is the discrete analogue of the continuous-time integrated variance:

$$IV = \int_0^T \sigma_t^2 \, dt$$

which appears in **Itô's lemma** and underlies the Black–Scholes PDE. Under the semi-martingale assumption, the realized variance converges to the quadratic variation $[X]_T$ as the sampling frequency increases (Barndorff-Nielsen & Shephard, 2002).

### 3.2 Supervised Regression Framing

The constructed dataset is cast as a standard supervised regression problem: given window-level microstructure summary statistics, predict $\sigma^{RV}$. The modelling choices at this stage are deliberately framework-agnostic—the pipeline is compatible with linear regression, gradient boosting (LightGBM/XGBoost), and neural networks. The Kaggle benchmark uses RMSPE as the loss metric:

$$\mathrm{RMSPE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{y_i}\right)^2}$$

## 4. FRM Concepts Applied

This project maps directly to multiple topics from the **FRM Part I and Part II** curriculum:

| FRM Topic | Book / Part | Application in Project |
|---|---|---|
| Log-returns & Price Dynamics | Part I — Foundations | WAP log-return series $r_t$ |
| Realized Variance / GARCH | Part II — Market Risk | $\sigma^{RV}$ as high-frequency vol |
| Bid-Ask Spread & Liquidity Risk | Part II — Liquidity Risk | Spread $S_t$ as liquidity proxy |
| Order Book & Market Microstructure | Part II — Current Issues | LOB feature engineering |
| Value-at-Risk Foundations | Part I — Quantitative | Log-normality of returns assumed |
| Black–Scholes & Quadratic Variation | Part II — Valuation | $IV = \int_0^T \sigma_t^2 dt$ |

The WAP formula itself reflects the **mid-price correction** recommended in the FRM curriculum's treatment of transaction cost analysis—using size-weighted rather than arithmetic mid to better approximate where the next trade will occur.

## 5. Interpretation of Results

**WAP as Fair Value:** By weighting prices against the *opposite side's* quantity, WAP is more stable than the arithmetic mid-price. A large bid-ask spread signals illiquidity and typically precedes elevated realized volatility—consistent with Amihud's illiquidity ratio and the FRM treatment of liquidity-adjusted VaR.

**Order-Flow Imbalance:** A strongly positive OFI (excess buy pressure) tends to predict short-term upward WAP movement and increased microstructure noise—both of which inflate realized volatility estimates. This is consistent with the *Kyle (1985)* lambda framework for price impact.

**Realized Volatility Behaviour:** Across the Optiver universe, $\sigma^{RV}$ is right-skewed and exhibits strong cross-sectional variation, consistent with the stylized facts of high-frequency returns documented in the FRM curriculum. The sum-of-squares estimator is robust under irregular tick spacing, making it preferable to OHLC-based estimators (e.g., Garman–Klass) for intraday data.

**Prediction Insight:** Stocks with persistently wide spreads and high imbalance variance tend to have less predictable realized volatility—a finding with direct implications for options market-making and delta-hedging frequency decisions.

## 6. Field and Type of Quant Trading

This project sits at the intersection of three areas of quantitative finance:

1. **High-Frequency Trading (HFT) / Market Making:** Realized volatility prediction is the core input to inventory risk management for market makers. A market maker quoting a two-sided spread needs to estimate short-horizon vol to set spreads wide enough to cover adverse selection risk.

2. **Derivatives Pricing & Volatility Trading:** The $\sigma^{RV}$ estimate underpins variance swap pricing and is compared against implied volatility ($\sigma^{IV}$) to identify richness/cheapness in options. The gap between $\sigma^{IV}$ and $\sigma^{RV}$ defines the *variance risk premium*—a systematically harvested factor in volatility hedge funds.

3. **Statistical / Systematic Alpha Research:** OFI and spread-based signals are well-documented short-horizon alpha factors (Cont, Kukanov & Stoikov, 2014). At lower frequencies, aggregated

microstructure signals feed into multi-factor equity models—precisely the use-case described in roles targeting NSE/BSE systematic strategies.

**Relevance to Indian Markets (NSE/BSE):** NSE's OPSTRA and raw F&O tick data provide equivalent LOB depth. The pipeline is directly portable to Nifty 50 and BankNifty options for intraday volatility signal generation, hedging cost estimation, and as a pre-processing step to the options alpha suite described in the author's broader research portfolio.

---

**Data:** Optiver Realized Volatility Prediction, Kaggle (2021) — https://www.kaggle.com/c/optiver-realized-volatility-predicti

**Key References:** Andersen & Bollerslev (1998); Barndorff-Nielsen & Shephard (2002); Cont, Kukanov & Stoikov (2014); Kyle (1985); GARP FRM Curriculum Parts I & II.